Ruiyang Hao

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Employment And Education

Institute for AI Industry Research, Tsinghua University, Beijing, China | 2023.7-now

Research Assistant | Supervisor: Prof. Zaiqing Nie

Research interest: Computer Vision for Autonomous Driving

SenseTime, Beijing, China | 2022.6-2022.10

Research Intern | Supervisor: Dr. Jiang Wu

Research interest: Computer Vision for UAV Remote Sensing

Department of Automation, Tsinghua University, Beijing, China | 2020.9-2023.7

Electronic Engineering (MPhil) | GPA 4.00 | ranking 1/159 | Supervisor: Prof. Biqing Huang

Research interest: Computer Vision for Industrial Inspection

Shenyuan Honors Collage, Beihang University, Beijing, China | 2016.9-2020.6

Electronic Engineering (BEng) | GPA 3.83 | ranking 1/51 | Supervisor: Prof. Ying Cheng

Research interest: Combinatorial Optimization for Industry

Research Experience

Big Data Intelligence Lab, Institute for AI Industry Research, Tsinghua | Supervisor: Prof. Zaiqing Nie

- **Research Topic**: Computer Vision for Autonomous Driving
- **Specific Research Projects**:
  - ✓ **Roadside Cooperative Perception (RCooper) Datasets and Benchmark**

    Objective: Towards high-level autonomous driving, **R**oadside **Coo**perative **Per**ception (RCooper) task is proposed to realize more practical area-coverage roadside perception for restricted traffic areas, which can extend sensing range and reduce blind spots.

    Contributions: We construct and release the **first real-world, large-scale RCooper dataset** to bloom the research on practical roadside cooperative perception, including detection and tracking. The manually annotated dataset comprises 50k images and 30k point clouds, including two representative traffic scenes (i.e., intersection and corridor). The constructed benchmarks prove the **effectiveness of roadside cooperation perception** and **demonstrate the direction of further research**.

  - ✓ **More Powerful End-to-End Autonomous Driving**

    Objective: We are building more powerful end-to-end autonomous driving model via self-supervised learning and robust planning methods.

Institute of Industrial Intelligence System, Department of Automation, Tsinghua | Supervisor: Prof. Biqing Huang

- **Research Topic**: Computer Vision for Industrial Inspection
- **Research Overview**: Automatic surface defect detection in industrial products plays a crucial role in quality control. Deep learning techniques have emerged as powerful tools for this task, building upon benchmark methods in object detection and semantic segmentation. To address the **challenge of limited data**, various strategies are employed, including data augmentation, self-supervised learning, and metric learning. Furthermore, convolution's variations, multi-scale feature fusion, and channel/spatial attention mechanisms enhance the capability to **extract and fuse complex defect features**. Additionally, ensemble learning and weight decay methods effectively tackle the **long-tail defect detection problem**. The efficacy of these models and algorithms is rigorously validated through extensive experiments.

- **Specific Research Projects**:
  - ✓ **Glass defect identification project towards Fuyao Group**

    Objective: To realize automatic detection and identification of glass surface defects.

    Methodology: The **adaptive threshold segmentation** method was used to carry out high-recall and high-efficiency preliminary screening of glass defects, and a **metric learning task** was designed to extract robust embedding features of glass defects under the condition of limited samples. An ensemble learning framework is constructed for glass defect recognition, where **multi-expert classifiers with diversified loss functions** are constructed to train to solve the accuracy loss caused by long-tail data distribution.

    Effect: Glass recognition accuracy reached 94% on the real data set (SOTA).

  - ✓ **Unsupervised defect segmentation project**

    Objective: To achieve accurate segmentation of industrial surface defects in an unsupervised way.

    Methodology: A **lightweight network structure** based on depth-wise convolution is designed for industrial environment deployment. Spital-level and channel-level **attention modules and multi-task strategies** are employed to improve the segmentation performance of defect edges. A **self-supervised learning strategy with homologous enhancement** is proposed to solve the defect segmentation problem without annotated information.

    Effect: Two typical industrial defect datasets with a pixel-level AUC of 98.40% (SOTA).

  - ✓ **Steel surface defect detection project**

    Objective: To achieve accurate detection of steel surface defects.

    Methodology: Based on the classical object detection framework Faster R-CNN, a novel architecture is proposed to address the issue of the diversity in defect shapes and scales: **deformable and atrous convolution** was adopted to improve the feature extraction towards multi-shape defects, and a **balanced feature pyramid with richer feature fusion** was designed to achieve better detection performance for multi-scale defects.

    Effect: 80.5% AP on steel surface defect datasets NEU-DET (SOTA).

SenseTime | Supervisor: Dr. Jiang Wu

- **Research Topic**: UAV remote sensing (semantic segmentation for solar panel plinth)
- **Research Summary**: To address the segmentation challenges posed by narrow and slender features of targets on solar panel plinths (e.g., roofs), the **parallel multi-resolution branch and multi-level feature fusion** capabilities of HRNet are leveraged, which effectively improves the segmentation performance of small targets. Additionally, A **multi-head multi-task decoder** is constructed to enhance the edge segmentation and direction perception. Furthermore, **multi-modality integration** is employed, incorporating gradient information from the depth map to refine the perception of depth information. Finally, a **post-processing method considering continuity** is implemented and the detected orientation of the principal normal vector is utilized to restore discontinuous targets. These interventions perform a significant increase of over 30% in foreground pixel recall for UAV segmentation.

Service-oriented Intelligent Manufacturing Research Group, Beihang University | Supervisor: Prof. Ying Cheng

- **Research Topic**: Combinatorial Optimization for Industry (Manufacturing service supply-demand optimization)
- **Research Summary**: The modern manufacturing industry realizes effective resource sharing and close cooperation among manufacturing enterprises in the form of manufacturing services. Oriented towards the type diversity and quantity variation diversity of manufacturing service supply and demand, a **classification hypernetwork model** is established to describe the matching relationship between supply and demand of manufacturing services and their quantity relationship, and a **utility optimization model** considering the type diversity and quantity diversity of supply and demand is proposed. The **knowledge-guided NSGA-II multi-objective optimization** algorithm is proposed to solve the above optimization model, and the effectiveness and feasibility of the model and algorithm are demonstrated by experiments, and suggestions are put forward for the management of manufacturing enterprise clustering.

PUBLICATIONS & PATENTS

- **Ruiyang Hao**, Bingyu Lu, Ying Cheng, Xiu Li, & Biqing Huang[†] (2021). A Steel Surface Defect Inspection Approach towards Smart Industrial Monitoring. Journal of Intelligent Manufacturing, 32(7): 1833-1843. (IF: 8.3, 75 citations)
- **Ruiyang Hao**, Ying Cheng[†], Yongping Zhang, & Fei Tao (2021). Manufacturing Service Supply-Demand Optimization with Dual Diversities for Industrial Internet Platforms. Computers & Industrial Engineering, 156: 107237. (IF: 7.9, 24 citations)
- Ronge Xu*, **Ruiyang Hao***, & Biqing Huang[†] (2022). Efficient surface defect detection using self-supervised learning strategy and segmentation. Advanced Engineering Informatics, 52: 101566. (***co-first author**, IF: 8.8, 28 citations)
- Biqing Huang[†], **Ruiyang Hao** (2022). Method and device for glass defect detection based on deformable convolution and feature fusion. China Patent: CN114170230B.
- Yani Wang, Xiang Wang, **Ruiyang Hao**, Bingyu Lu, & Biqing Huang[†] (2024). Metal surface defect detection method based on improved cascade R-CNN. Journal of Computing & Information Science in Engineering, 24(4): 041002. (IF: 3.1)
- **Ruiyang Hao*,** Siqi Fan*, Yingru Dai, Zhenlin Zhang, Chenxi Li, Yuntian Wang, Haibao Yu, Wenxian Yang, Jirui Yuan, & Zaiqing Nie[†] (2023). RCooper: A Real-world Large-scale Dataset for Roadside Cooperative Perception. Under review.

Notes: * denotes equal contribution and [†] denotes corresponding author.

PROFESSIONAL SERVICES

- Reviewer of *Advanced Engineering Informatics*, *Journal of Industrial Information Integration*, *Journal of Intelligent Manufacturing*, *Journal of Real-Time Image Processing*, etc.
- Student Assistant to the Master's Degree Defense Committee of Tsinghua University.

MAIN AWARDS

- **National Scholarship**, awarded by the Ministry of Education of the PRC, in 2018, **top 1%** of undergraduates
- First-class Excellence Scholarship, awarded by Tsinghua University, in 2022, **top 2.5%** of graduate students
- First-class Excellence Scholarship, awarded by Tsinghua University, in 2021, **top 2.5%** of graduate students
- Outstanding graduate and the honored degree, awarded by Beihang University, in 2020, **top 5%** of undergraduates
- Meritorious Prize of Interdisciplinary Contest in Modeling, awarded by the American Consortium for Mathematics and Its Applications, in 2019, **top 8%** of contestants
- Meritorious Prize of Interdisciplinary Contest in Modeling, awarded by the American Consortium for Mathematics and Its Applications, in 2018, **top 8%** of contestants